

7.2 Orsat apparatus : percentage of CO_2

Describe the apparatus that is commonly used for the analysis of dry exhaust gases. Discuss the probable sources of error in such an analysis.

A boiler burns coal of the following composition : $\text{C} = 88\%$, $\text{H}_2 = 3.8\%$, $\text{O}_2 = 2.2\%$ and the remainder ash. On a particular occasion the percentage of CO_2 passing up the chimney was 10%. The temperature of chimney gases was then 250°C . If sample of this gas is analysed by the Orsat apparatus at room temperature, what percentage of CO_2 would you expect, assuming complete combustion of the fuel.

Solution. For theory— see text.

Total exhaust gases = $\text{CO}_2 + \text{H}_2\text{O} + \text{O}_2 + \text{N}_2 + \text{CO}$

$$\text{Actual percentage of } \text{CO}_2 = \frac{\text{CO}_2 \times 100}{\text{CO}_2 + \text{H}_2\text{O} + \text{O}_2 + \text{N}_2}$$

Handwritten notes on the right margin:

$$\begin{array}{r} \text{CO}_2 \\ \text{CO} \\ \text{O}_2 \\ \hline \text{N}_2 \\ \hline \end{array}$$

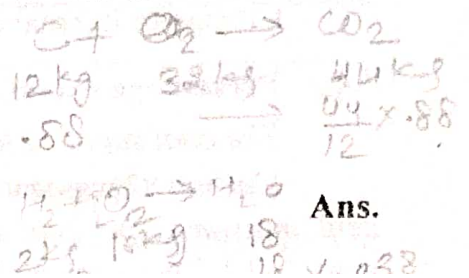
Illustrative examples

Mass of constituents per kg of fuel (a)	Mass of products of combustion (b)		Molar mass (c)		Proportional volume (d) = (b)/(c)	
	CO ₂	H ₂ O	CO ₂	H ₂ O	CO ₂	H ₂ O
C = 0.880 H ₂ = 0.038 O ₂ = 0.022	$0.88 \times \frac{44}{12}$	0.038×9	44	18	0.0733	0.019

$$10 = \frac{0.0733 \times 100}{0.0733 + 0.019 + O_2 + N_2}, \therefore O_2 + N_2 = 0.641$$

In Orsat apparatus only dfg is measured

$$\therefore \text{Percentage of CO}_2 = \frac{CO_2 \times 100}{CO_2 + O_2 + N_2} = \frac{0.0733 \times 100}{0.0733 + 0.641} = 10.28\%$$



Ans.

7.4 Percentage excess air ; c_p , given fuel and dfg analysis

The coal supplied to a boiler furnace has the following ultimate analysis :

Carbon 82 per cent, hydrogen 5.4 per cent, oxygen 7 per cent, nitrogen 0.5 per cent, sulphur 0.6 per cent, moisture 1 per cent and the rest is ash. The volumetric analysis of the dry flue gases is CO_2 9.6 per cent, CO 1.2 per cent, N_2 81.4 per cent and O_2 7.8 per cent. Determine,

(a) percentage of excess air supplied to the boiler furnace (b) mean specific heat of the dry flue gases.

Given the specific heats at constant pressure ; $\text{CO}_2 = 0.879$, $\text{O}_2 = 0.913$, $\text{N}_2 = 1.022$ and CO = 1.038 in kJ/kg K units.

Solution. (a) Mass of air required per kg of fuel for complete combustion

$$= \frac{100}{23} \left[0.82 \times \frac{8}{3} + 0.054 \times 8 + 0.06 \times 1 - 0.07 \right] = 11.11 \text{ kg}$$

Volume of constituents per mol of dfg (a)	Molar mass (b)	Proportional mass of constituents (c) = (a) × (b)	Mass per kg of flue gas (d) = (c)/Σ(c)	Mass of carbon per kg of flue gas (e)
$\text{CO}_2 = 9.6$	44	$9.6 \times 44 = 422.4$	0.14160	$\frac{0.1416 \times 12}{44} = 0.03862$
$\text{CO} = 1.2$	28	$1.2 \times 28 = 33.6$	0.01126	$\frac{0.01126 \times 12}{28} = 0.00482$
$\text{N}_2 = 81.4$	28	$81.4 \times 28 = 2279$	0.76350	
$\text{O}_2 = 7.8$	32	$7.8 \times 32 = 249.6$	0.08364	
Total = 100.0		2984.6	1.00000	0.04344

$$\text{Mass of dry flue gases per kg of fuel} = \frac{\text{mass of carbon per kg of fuel}}{\text{mass of carbon per kg of fuel}} = \frac{0.82}{0.04344} = 18.88 \text{ kg}$$

$$\text{From table above : Mass of CO per kg fuel} = 18.88 \times 0.01126 = 0.2126 \text{ kg}$$

$$\text{Mass of unused } \text{O}_2 \text{ per kg of fuel} = 18.88 \times 0.08364 = 1.5791 \text{ kg}$$

$$\text{Mass of } \text{O}_2 \text{ required for burning of CO} = \frac{0.2126 \times 4}{7} = 0.1215 \text{ kg}$$

$$\text{Excess } \text{O}_2 \text{ per kg of fuel} = 1.5791 - 0.1215 = 1.4576 \text{ kg}$$

$$\text{Excess air} = \frac{1.4576 \times 100}{23} = 6.3374 \text{ or Percentage excess air} = \frac{6.3374}{11.11} \times 100 = 57.05\% \text{ Ans.}$$

(b) Mean specific heat of dry flue gases

Constituents (a)	Mass per kg of dry flue gas (b)	Specific heat of constituents at constant pressure (c)	Heat per kg of dry flue gas (d) = (b) × (c)
CO ₂	0.14160	0.879	0.1245
CO	0.01126	0.913	0.0103
N ₂	0.76350	1.022	0.7803
O ₂	0.08364	1.038	0.0868
Total	1.00000		1.0019

Mean value of specific heat of dry flue gas, $c_p = 1.0019 \text{ kJ/kg K}$

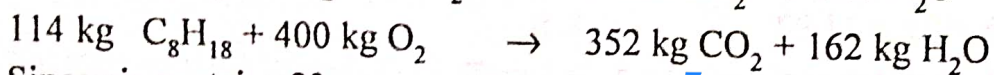
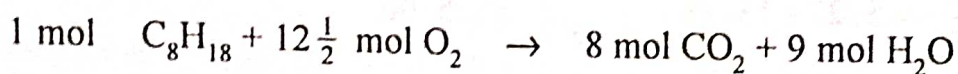
Ans.

7.9 Stoichiometric Air/fuel ratio ; mass and volumetric analysis of products

A fuel having formula C_8H_{18} is burnt with chemically correct amount of air at atmospheric pressure. Write the reaction equation and find the stoichiometric air fuel ratio. Also calculate the percentage by mass and volume of products at atmospheric pressure.

Solution. Since the combustion is theoretically complete with the amount of stoichiometric air the products of combustion will contain only CO_2 and H_2O and N_2 brought with the air. The combustion equation is written. The amount of O_2 required is found by first balancing the C and H atoms on each side of the equation and then summing the number of atoms of O in the products. The amount of H_2 is then amount associated with the O_2 supplied from the air. N_2 does not take part in the combustion, so will remain same, before and after combustion.

Reaction equation is thus

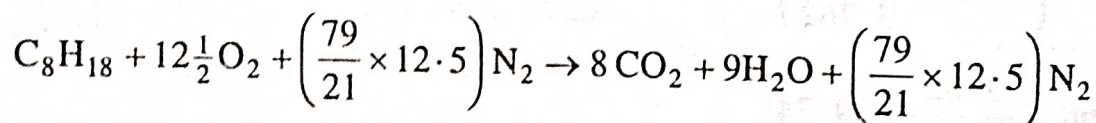


Since air contains 23 percent of O_2 by mass

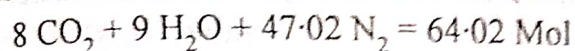
$$\text{Stoichiometric air/fuel ratio} = \frac{400 \times 100}{114 \times 23} = 15.256$$

Ans.

Reaction equation considering N_2 supplied along with O_2 from air is thus



The constituents of products of combustion (in mol) are



We know that the mol fraction of a product equals the volume fraction. Hence volume percentage of constituents in products are :

$$\text{CO}_2 = \frac{8}{64.02} \times 100 = 12.50\% ; \text{H}_2\text{O} = \frac{9}{64.02} \times 100 = 14.06\% ; \text{N}_2 = \frac{47.02}{64.02} \times 100 = 73.44\% \quad \text{Ans.}$$

The volumetric composition can be converted to mass composition as shown

Volumetric composition in products (a)	Molecular mass of constituents (b)	Mass composition in products (c) = (a) × (b)	Mass percentage (d) = (c)/Σ(c) × 100
CO ₂ = 12.60	44	550	19.24
H ₂ O = 14.06	18	253	8.85
N ₂ = 73.44	28	2056	71.91
		Σ (c) = 2859	100.00